TecniZoneApp(Classes Modeling)

ATTRIBUTES AND METHODS OF EACH CLASS

Class SystemUser

Attributes:
- username : String
- password : String
- userRole : String

Methods:
+ login(passwordEntered : String) : boolean
+ updatePassword(newPassword : String)

Class Client

Attributes:
- clientId : String
- fullName : String
- phoneNumber : String
- companyName : String

Methods:
+ updateContactInfo(newPhone : String)
+ registerEquipment(deviceDetails : String) : Equipment

Class Technician

Attributes:
- technicianId : String
- fullName : String
- specialty : String

Methods:
+ updateSpecialty(newSpecialty : String)
+ assignToOrder(order : RepairOrder)

Class Supplier

Attributes:
- supplierId : String
- companyName : String
- contactPhone : String

Methods:
+ orderParts(partNeeded : Part, quantity : int)

Class Part

Attributes:
- partId : String
- partName : String
- stockQuantity : int

- minimumStock : int
- unitPrice : double
Methods:
+ reduceStock(quantityUsed : int)
+ addStock(quantityReceived : int)
- checkMinimumStock() : StockAlert

Class ServiceCatalog
Attributes:
- serviceCode : String
- serviceName : String
- fixedLaborCost : double
Methods:
+ updateCost(newCost : double)

Class Equipment
Attributes:
- serialNumber : String
- deviceType : String
- brand : String
- model : String
- reportedFaults : String
- ownerClient : Client
Methods:
+ updateReportedFault(newFault : String)
+ createWarrantyCheck() : WarrantyChecklist

Class WarrantyChecklist
Attributes:
- checklistId : String
- relatedEquipment : Equipment
- hasPhysicalDamage : boolean
- includesAccessories : boolean
- isWarrantyValid : boolean
Methods:
+ markPhysicalDamage(description : String)
+ validateWarrantyStatus() : boolean

Class RepairOrder
Attributes:
- orderNumber : String
- customer : Client
- device : Equipment
- assignedTech : Technician

- currentStatus : String
- estimatedDeliveryDate : Date
- laborCost : double
- partsCost : double
- totalAmount : double
Methods:
+ assignTechnician(tech : Technician)
+ addPartToOrder(part : Part, quantity : int) :
OrderPartDetail
+ changeStatus(newStatus : String, user : SystemUser) :
StatusLog
+ calculateTotalCost()
+ generateInvoice() : Invoice

Class StatusLog
Attributes:
- logId : String
- order : RepairOrder
- statusChangedTo : String
- changeTimestamp : DateTime
- changedByUser : SystemUser
Methods:
+ recordChangeTimestamp()

Class OrderPartDetail
Attributes:
- detailId : String
- order : RepairOrder
- usedPart : Part
- quantityUsed : int
- subtotal : double
Methods:
+ calculateSubtotal() : double

Class StockAlert
Attributes:
- alertId : String
- affectedPart : Part
- triggerDate : DateTime
- alertMessage : String
- isResolved : boolean
Methods:
+ markAsResolved(user : SystemUser)

Class DirectSale
Attributes:
- saleId : String
- saleDate : DateTime
- totalAmount : double
Methods:
+ sellPart(part : Part, quantity : int)
+ generateInvoice() : Invoice

Class Invoice
Attributes:
- invoiceNumber : String
- repairOrder : RepairOrder
- issueDate : DateTime
- finalTotal : double
- paymentMethod : String
Methods:
+ processPayment(paymentMethod : String)
+ printReceipt()

Class MonthlyReport
Attributes:
- reportId : String
- monthAndYear : String
- totalCompletedRepairs : int
- totalMoneyCollected : double
Methods:
+ compileMonthlyData(month : String, year : int)
+ exportToText()

https://docs.google.com/document/d/1CrG1caZp_V0aO7stfjrCGYK56W3A6qGZg43KQThd93E/edit?usp=sharing